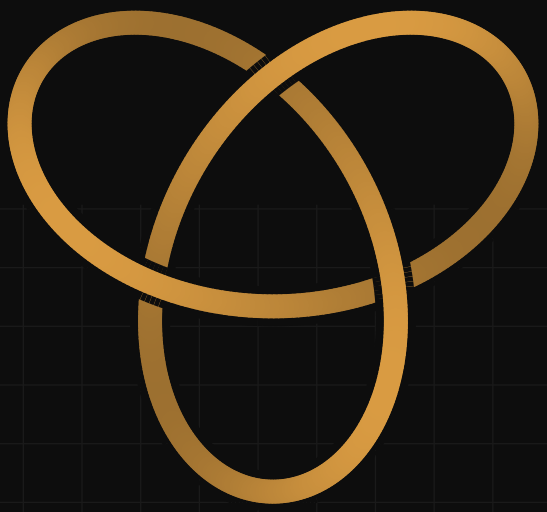
AKSIOMA

TECHNICAL GUIDE

The .aknot file format

How AKSIOMA Knots saves a knot, opens it again, and teaches Windows to recognise the file.



JSON	**v1**	**AKSIOMA.Knot**	**4**
plain UTF-8 text	format version	Windows file type	registry entries

AKSIOMA Knots 1.0.0 · github.com/Albionn1/AKSIOMA-Knots

OVERVIEW

What an .aknot file is

An .aknot file is a small text file that remembers one knot exactly as you left it in AKSIOMA Knots: which knot it is, the shape of its curve and how the tube looked.

Inside, it is **JSON**, the same text format web pages use to exchange data. You can open one in Notepad and read it. The program writes it when you choose `File › Save knot...` (Ctrl+S) and reads it when you choose `File › Open knot...` (Ctrl+O) or double-click the file in Explorer.

Every file stores the knot **twice**: once as a *recipe* ("the trefoil", "torus knot T(5,2)", "knot 8_19 from the table") and once as the *actual list of 3D points* on screen. The recipe rebuilds the knot perfectly; the points keep shapes the recipe cannot describe, such as a knot you relaxed or scrambled, and let any other program read the curve.

	Contents	
1	What an .aknot file is	2
2	Inside a file: a real example	3
3	Every field explained	4
4	Knot types and the two copies of the knot	5
5	What happens when you save	6
6	What happens when you open	7
7	Other files it can open, and error messages	8
8	How Windows recognises .aknot files	9
9	From double-click to knot, and good to know	10

At a glance

PROPERTY	VALUE
Extension	`.aknot` (the save dialog adds it if you leave it out)
Content	JSON text, UTF-8, indented with 4 spaces
Format marker	`"format": "AKSIOMA Knots"`, `"version": 1`
Size	Tens of kilobytes, almost all of it the list of points
Code	`knotexport.cpp` (read and write), `mainwindow.cpp` (save and open), `appsettings.cpp` (Windows file type)
Windows file type	`AKSIOMA.Knot`, "AKSIOMA knot file" / "Skedar nyjeje AKSIOMA"

SECTION 2

Inside a file: a real example

This is what a saved trefoil looks like when you open it in a text editor. Only the list of points is shortened.

Trefoil Knot.aknot


```
{
        "appearance": {
            "color": "#d99b42",
            "colorMode": 0,
            "material": 0,
            "samples": 160,
            "shine": 0.7,
            "thickness": 0.16
        },
        "format": "AKSIOMA Knots",
        "knot": {
            "custom": "",
            "p": 3,
            "q": 2,
            "table": "",
            "type": "trefoil"
        },
        "name": "Trefoil Knot",
        "points": [
            [
                0.39132802580830667,
                -0.9784593474571476,
                -0.2334453638559054
            ],
            [
                0.1961780072147585,
                -0.994605631225533,
                -0.11753739745783764
            ],
            ...   (one group per point)
        ],
        "shapeChanged": false,
        "version": 1
    }
```


appearance
How the tube looked: the sliders and menus of the STYLE tab.

format, version
The signature. Without it the program refuses the file.

knot
The recipe: which knot, and its settings.

name
The title shown in the program.

points
The curve itself, as x, y, z coordinates.

shapeChanged
true if the knot was relaxed or scrambled.

Why the keys are in this order
Qt writes the keys in **alphabetical order** and puts every number of a point on its own line, which is why the file looks longer than the short form in the code comment (`knotexport.h`). The order does not matter when reading: any JSON reader finds the keys by name.

SECTION 3

Every field explained

Three groups: the signature and name at the top level, the recipe in “knot”, and the look in “appearance”.

Top level

KEY	TYPE	MEANING
`format`	text	Always `"AKSIOMA Knots"`. Any other value: “This is not an AKSIOMA Knots file.”
`version`	number	Format version, now `1`. A higher number: “The file was saved by a newer version of AKSIOMA Knots.”
`name`	text	Title of the knot. Missing: the file name is used.
`shapeChanged`	true/false	The curve was moved by Relax or Scramble, so the recipe no longer describes it and the points are used instead.
`points`	list	The curve on screen: `[[x, y, z], ...]`. Groups that do not have exactly three numbers, or contain infinite values, are skipped.
`knot`	object	The recipe (below).
`appearance`	object	Optional. Missing: the current look in the program is kept.

knot

KEY	TYPE	MEANING
`type`	text	Which knot: one of the ten keys on the next page.
`p, q`	whole number	Torus knot T(p,q). Kept between 1 and 20 when read. Saved for every type (default 3 and 2).
`table`	text	KnotInfo name for `"table"`, e.g. `"8_19"`, `"11n_34"`. Empty otherwise.
`custom`	text	For `"custom"`: the formula or coordinate text exactly as typed in the Custom Curve box. Empty otherwise.

appearance

KEY	STORED AS	IN THE PROGRAM
`thickness`	0.02 – 0.60	Tube thickness slider (slider value ÷ 100); default 0.16
`samples`	48 – 320	Curve density slider; default 160
`material`	0 – 3	0 Glossy, 1 Satin, 2 Metal, 3 Rope
`shine`	0.00 – 1.00	Material shine slider (value ÷ 100)
`colorMode`	0 – 2	0 Single Color, 1 Crossing Colors, 2 Strand Colors
`color`	#rrggbb	Tube colour, e.g. `"#d99b42"` (the AKSIOMA amber)

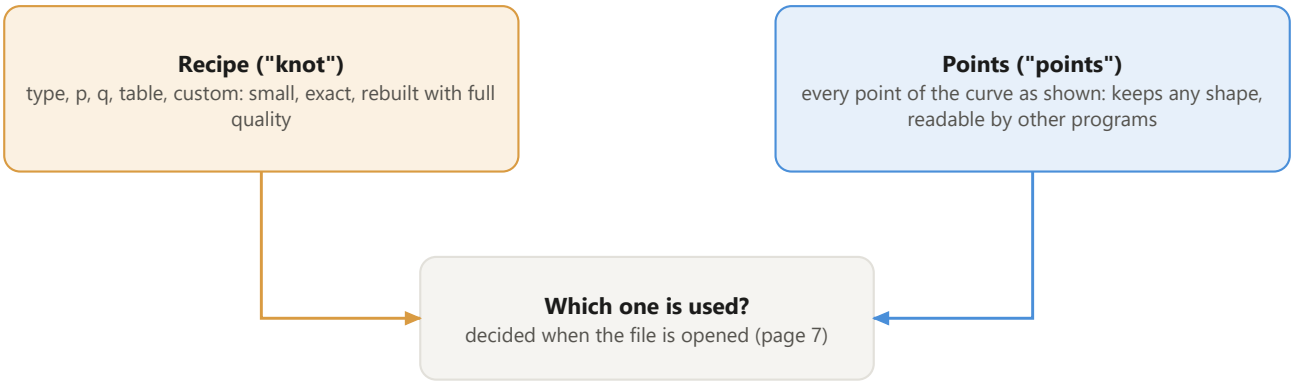
SECTION 4

Knot types and the two copies of the knot

The “type” key uses fixed English words, so a file saved in the Albanian program opens in the English one and the other way round.

TYPE	KNOT	REBUILT FROM
`unknot`	Unknot (a plain circle)	the recipe
`trefoil`	Trefoil knot 3_1	the recipe
`figure-eight`	Figure-eight knot 4_1	the recipe
`cinquefoil`	Cinquefoil knot 5_1	the recipe
`stevedore`	Stevedore knot 6_1	the recipe
`dna`	DNA double helix (two strands with rungs)	the recipe
`celtic`	Celtic chain of four rings	the recipe
`torus`	Torus knot T(p,q)	the recipe with `p` and `q`
`table`	A knot from the Knot Browser (12,965 knots)	the KnotInfo name in `table`
`custom`	Your own formula or coordinates	the text in `custom`

Why store the knot twice?



- **Normally the recipe wins.** A trefoil is rebuilt from its formula, so it comes back perfectly smooth at whatever curve density you choose, and the knot keeps its name and everything the program knows about it.
- **The points take over** when the recipe cannot be trusted: the knot was relaxed or scrambled (`shapeChanged: true`), the type is a word this version does not know, or the table name is not in its knot table.
- **Other programs** (Python, MATLAB, Blender scripts) only need the points: they can draw or analyse the curve without knowing anything about AKSIOMA.

SECTION 5

What happens when you save

`MainWindow::saveKnotFile()` collects the current state, then `KnotExport::writeKnotFile()` turns it into JSON and writes it.

1. **Stop the motion**
   A running Relax or Scramble is paused, so the saved points are the shape you see.
2. **Collect the recipe**
   Knot type, p and q, the table name or the custom text.
3. **Collect the curve**
   The points of the curve on screen (for DNA and the Celtic chain: the first strand) and the shapeChanged flag.
4. **Collect the look**
   Thickness, curve density, material, shine, colour mode, colour.
5. **Ask for a name**
   The dialog suggests the knot's name in the last folder you used; .aknot is added if the name has no extension.
6. **Write safely**
   QSaveFile writes to a temporary file and swaps it in only when everything is written.

Where each control ends up

IN THE PROGRAM	IN THE FILE
Knot menu (presets, T(p,q))	`knot.type`
Torus p and q	`knot.p`, `knot.q`
Knot chosen in the Knot Browser	`knot.table`
Custom Curve text box (also drawn knots)	`knot.custom`
The curve on screen, after Relax / Scramble too	`points`, `shapeChanged`
STYLE tab: thickness, density, material, shine, colours	`appearance`

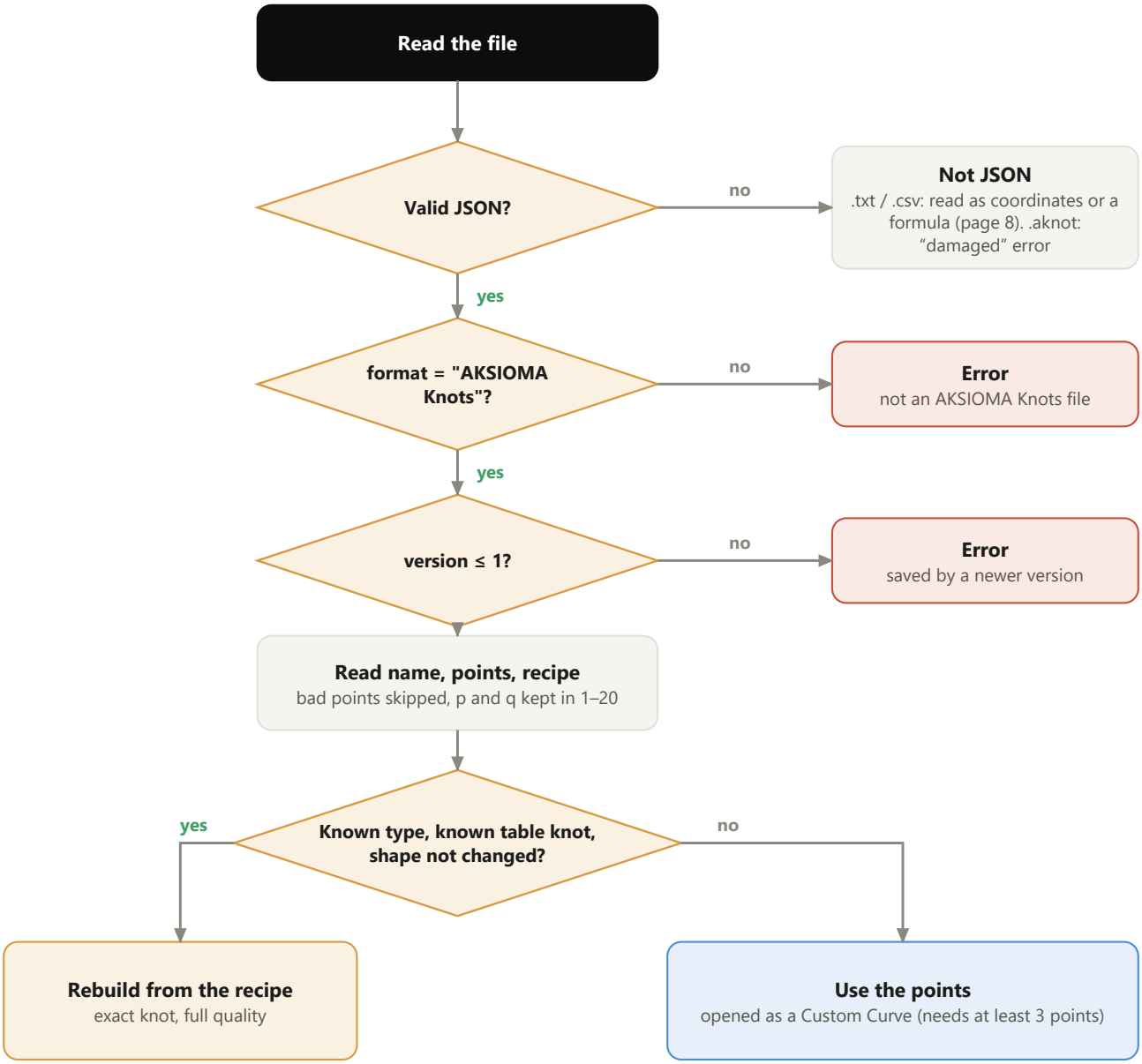
Why saving cannot damage an existing file
Because of the safe write in step 6, a crash or a full disk in the middle of saving never leaves a half-written file: you either get the new file or the old one stays untouched.

Small details
Nothing is saved when no knot is shown. The folder you save to is remembered (`AppSettings::setLastFolder`) and offered next time for opening, saving and exporting.

SECTION 6

What happens when you open

`KnotExport::readKnotFile()` checks the file step by step and decides whether the knot is rebuilt from its recipe or from its points.



After that, the main window applies the file in this order: the look first (thickness, density, material, shine, colour mode, colour), then p and q, the table name or the custom text, then the knot choice. It rebuilds the knot **once**, updates the LEARN tab and resets the camera.

> **Why the order matters**
> The material is set before the shine because choosing a material also moves the shine slider to that material's usual value. Setting the shine afterwards restores the value that was saved.

SECTION 7

Other files it can open, and error messages

The Open dialog also accepts plain text and CSV files, so knots from other programs, spreadsheets or papers can be brought in.

A list of coordinates (.txt, .csv)

One point per line. Lines that do not start like a number (headers, `#` comments, empty lines) are skipped, and numbers may be in brackets.

spiral.csv

```
# my knot, exported from Python
    x, y, z
    1.000, 0.000, 0.000
    0.809, 0.588, 0.250
    (0.309, 0.951, 0.000)
    ...
```

A formula

If the text contains an `=` sign it is read as a Custom Curve formula in `t`, the same language as the Custom Curve box:

trefoil.txt

```
x = sin(t) + 2*sin(2*t); y = cos(t) - 2*cos(2*t); z = -sin(3*t)
```

Both are opened as a Custom Curve named after the file. An .aknot file is never read this way: if it is not valid JSON it is reported as damaged.

Error messages

MESSAGE	CAUSE	WHAT TO DO
The file is damaged (…)	An .aknot file that is not valid JSON, e.g. cut short or edited by hand	Open it in Notepad and look near the position given in brackets
This is not an AKSIOMA Knots file.	JSON without `format: AKSIOMA Knots`	Check the file was saved by AKSIOMA Knots
The file was saved by a newer version of AKSIOMA Knots.	version is higher than 1	Install the newest AKSIOMA Knots
The file contains no curve.	The points are needed but there are fewer than 3	Save the knot again
The file contains no coordinates (…)	A text file with no line that starts like a number	One “x, y, z” per line

All messages appear in a box titled “The knot could not be opened.”, in English or Albanian depending on the program's language.

SECTION 8

How Windows recognises .aknot files

Windows decides an icon and a program for every extension by looking it up in the registry. AKSIOMA Knots writes four entries there.

```
Registry

HKEY_CURRENT_USER\Software\Classes          (or HKEY_LOCAL_MACHINE for all users)
    │
    ├─ .aknot                          (Default) = AKSIOMA.Knot
    │
    └─ AKSIOMA.Knot                    (Default) = AKSIOMA knot file
       ├─ DefaultIcon                  (Default) = "C:\...\AKSIOMAKnots.exe",0
       └─ shell\open\command           (Default) = "C:\...\AKSIOMAKnots.exe" "%1"
```

ENTRY	WHAT WINDOWS DOES WITH IT
`.aknot`	Links the extension to the file type `AKSIOMA.Knot` (a "ProgID").
`AKSIOMA.Knot`	The text in Explorer's Type column: "AKSIOMA knot file", or "Skedar nyjeje AKSIOMA" in Albanian.
`DefaultIcon`	The file icon: icon number `0` inside AKSIOMAKnots.exe, the program's own icon (from `app.rc`).
`shell\open\command`	The command for a double click. Windows replaces `%1` with the full path of the file.

Two places set this up

1. The installer

Writes the entries for the person installing, or for all users if installed for everyone (Inno Setup HKA).
Uninstalling removes them again.
Tells Explorer to refresh (ChangesAssociations).

2. The program itself, at every start

Writes the same entries for the current user (AppSettings::registerFileType) if they differ, so a moved or unzipped copy still works.
Then calls SHChangeNotify: new icon at once.

The copy started last wins
Whichever copy of the program was started last owns the .aknot files. If you keep an old copy somewhere and start it, double-clicks open that copy until the installed one is started again.

SECTION 9

From double-click to knot, and good to know



The path is opened with a zero-delay timer, so the main window is fully on screen before the file is read.

Good to know

- **Language-independent.** Keys and type words are English; the name inside the file is in whatever language the knot was saved in.
- **Each double-click starts a new window.** The program does not hand the file to a copy that is already running.
- **Dragging a file onto the window** does not open it; use File › Open or a double-click.
- **Multi-strand structures** (DNA, Celtic chain) are rebuilt from the recipe; the points hold only the first strand.
- **Hand-editing is safe** as long as the JSON stays valid: change the colour, the name or the thickness in Notepad and open the file again.
- **Future versions** can add keys freely; version 1 ignores keys it does not know. Only a change that old versions would misread needs version 2.

Reading an .aknot file in Python

read_aknot.py

```
import json

    with open("Trefoil Knot.aknot", encoding="utf-8") as f:
        knot = json.load(f)

    assert knot["format"] == "AKSIOMA Knots"
    print(knot["name"], knot["knot"]["type"], len(knot["points"]), "points")

    xs, ys, zs = zip(*knot["points"])      # ready for matplotlib, numpy, Blender ...
```